

Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

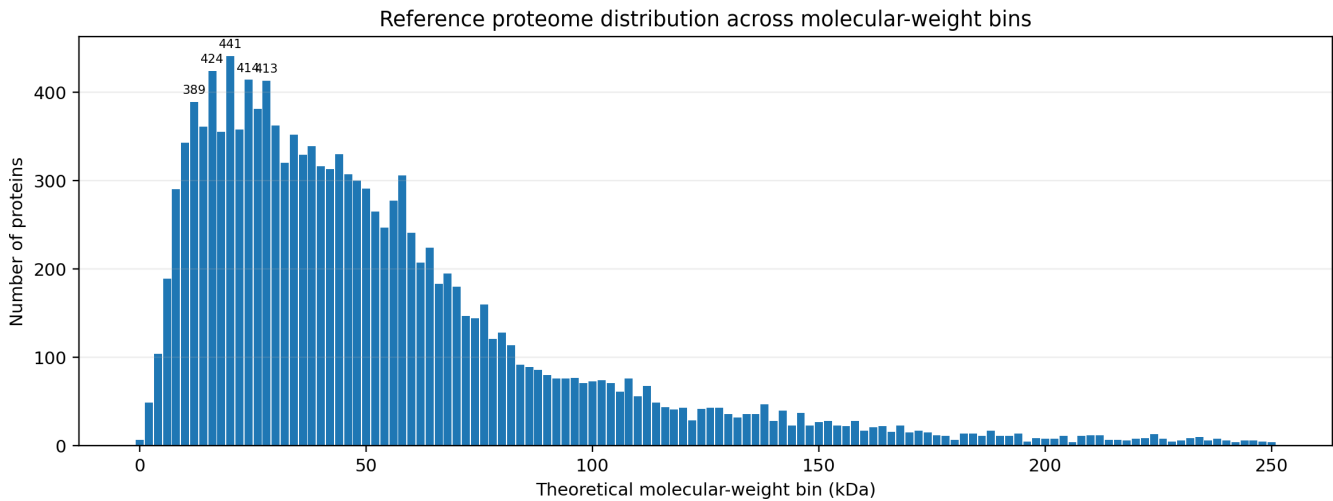
Metric	Value
Proteins parsed	13,817
Molecular-weight bins	232
Bin width	2 kDa
MW range	1.23 to 2443.39 kDa
Median MW	44.39 kDa
Mean MW	59.68 kDa
Proteins <= 50 kDa	7,786 (56.4%)
Proteins 50-100 kDa	4,077 (29.5%)
Proteins > 100 kDa	1,954 (14.1%)
High-priority bins	24
Elevated-priority bins	34
Routine-priority bins	174
Source FASTA	UP000000803_7227.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

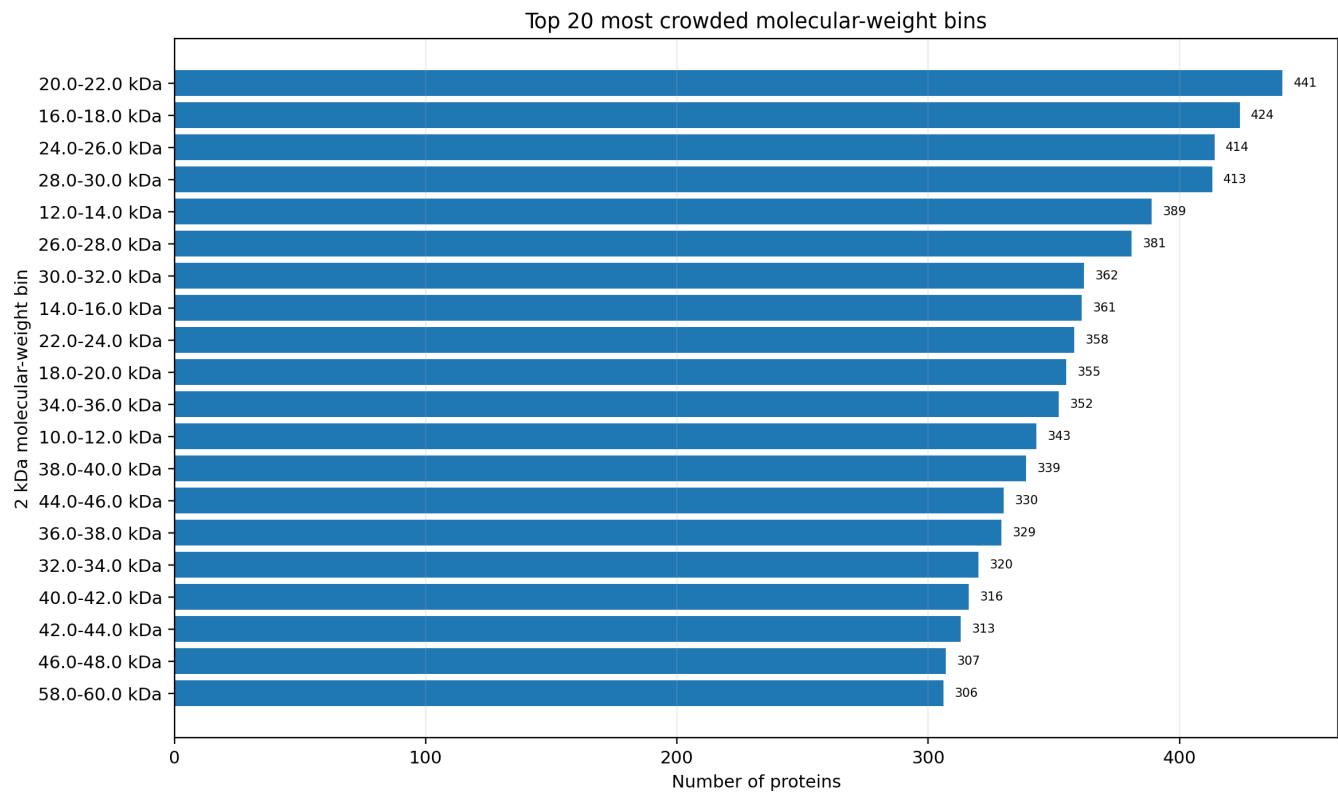
Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



Most crowded molecular-weight neighborhoods.

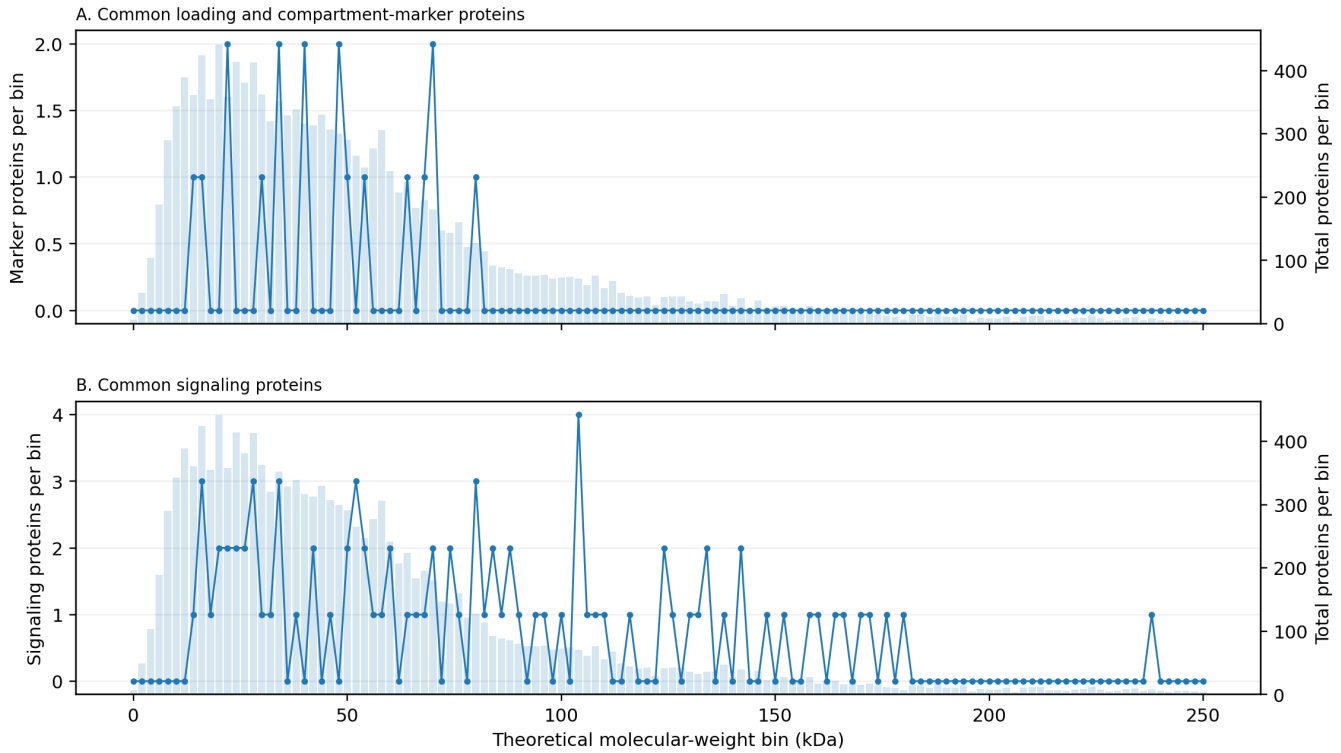
The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.



Distribution of common western blot markers and signaling proteins across molecular-weight bins.

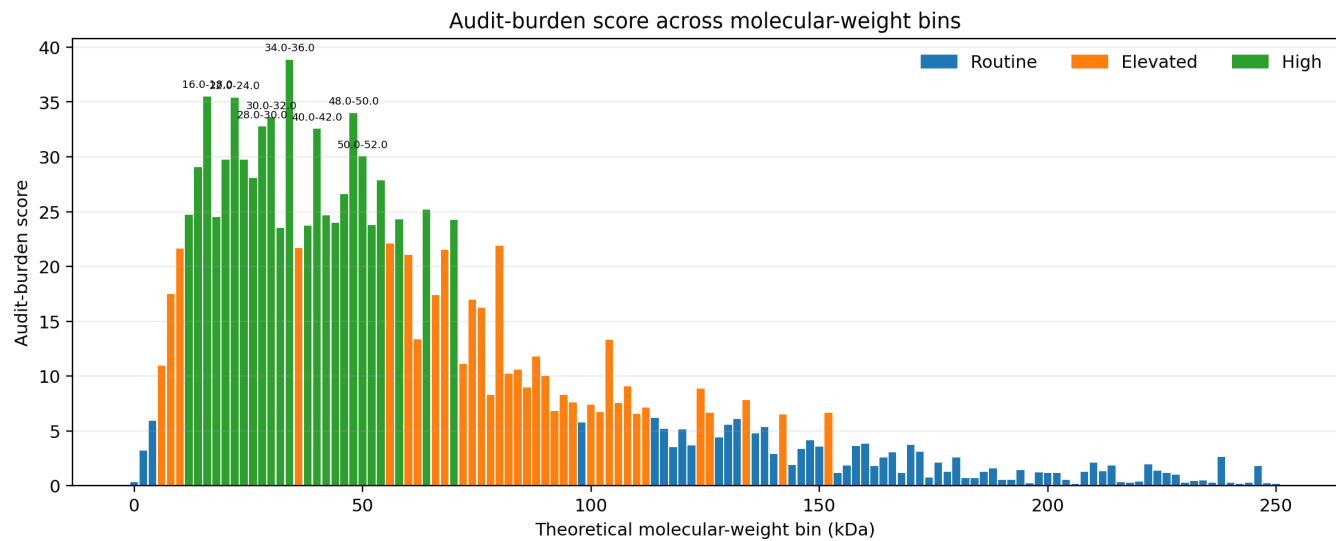
The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.

Audit-relevant annotations across molecular-weight bins



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
34.0-36.0 kDa	352	2	3	high	38.9
16.0-18.0 kDa	424	1	3	high	35.5
22.0-24.0 kDa	358	2	2	high	35.4
48.0-50.0 kDa	300	2	0	high	34.0
30.0-32.0 kDa	362	1	1	high	33.6
28.0-30.0 kDa	413	0	3	high	32.8
40.0-42.0 kDa	316	2	0	high	32.5
50.0-52.0 kDa	291	1	2	high	30.1
20.0-22.0 kDa	441	0	2	high	29.8
24.0-26.0 kDa	414	0	2	high	29.8
14.0-16.0 kDa	361	1	1	high	29.1
26.0-28.0 kDa	381	0	2	high	28.1
54.0-56.0 kDa	247	1	2	high	27.9
46.0-48.0 kDa	307	0	1	high	26.6
64.0-66.0 kDa	224	1	1	high	25.2
12.0-14.0 kDa	389	0	0	high	24.7
42.0-44.0 kDa	313	0	2	high	24.6
18.0-20.0 kDa	355	0	1	high	24.5
58.0-60.0 kDa	306	0	1	high	24.3
70.0-72.0 kDa	180	2	2	high	24.2
44.0-46.0 kDa	330	0	0	high	24.0
52.0-54.0 kDa	265	0	3	high	23.8
38.0-40.0 kDa	339	0	1	high	23.7
32.0-34.0 kDa	320	0	1	high	23.5
56.0-58.0 kDa	277	0	1	elevated	22.1

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
34.0-36.0 kDa	352	2	3	Actin_tubulin; Apoptosis; GAPDH_metabolic_controls; Notch; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome; Wnt_Wingless	high	TORIP, DmelCG10973, Teh1, DmelCG11617, RplL5 [Ribosomal_proteins], DmelCG34444, CAH14, LPP-like, DmelCG6067, DmelCG10465, MET30, DmelCG46434, DmelCG6972, DmelCG32371, Surf1, jhamt, DmelCG14420, Osi18
16.0-18.0 kDa	424	1	3	Actin_tubulin; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Notch; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	high	Chrac-16, DmelCG13689, DmelCG13239, DmelCG7781, Rbp1, RpS23 [Ribosomal_proteins], Rpl32 [Ribosomal_proteins], Npc2a, ND-B14.5AL [Mitochondrial_membrane], Rpl28 [Ribosomal_proteins], DmelCG9568, DmelCG11663, Pbars, DmelCG11300, MED21, DmelCG14658, Blos3, BcDNA:RE30174
22.0-24.0 kDa	358	2	2	Actin_tubulin; Heat_shock_chaperones; Mitochondrial_membrane; RAS_ERK_MAPK; Ribosomal_proteins; Ubiquitin_proteasome	high	Cib2, DmelCG31109, DmelCG11226, CG14286, DmelCG14495, ND-SGDH [Mitochondrial_membrane], BcDNA:GH11730, DmelCG16813, VhaPPA1-1, CG15250, DmelCG42830, DmelCG14455, BP1005, DmelCG32797, Rab18, DmelCG14777, DmelCG34138, DmelCG3955
48.0-50.0 kDa	300	2	0	Actin_tubulin; Apoptosis; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ubiquitin_proteasome; Wnt_Wingless	high	GalT5, pre-Ira-G, l(3)neo38, anon-WO0118547.264, PolD2 [Notch], sut4, rho-4, DmelCG10376, Diap1 [Apoptosis], CG13770, DmelCG3635, DmelCG10559, carboxypeptidase, 6959, Elg71Ea, cpt, M1BP, brwl
30.0-32.0 kDa	362	1	1	Actin_tubulin; Apoptosis; Autophagy; GAPDH_metabolic_controls; Histones; JNK_MAPK; Mitochondrial_membrane; Notch; Protein_phosphatases; Ribosomal_proteins; Toll_lmd_NFKB; Ubiquitin_proteasome	high	Fire-like, DmelCG10053, DmelCG13408, Jhbp5, BcDNA:SD21514, DmelCG13996, lp259, Fst, Ric, Osi7, Q9VXA9, fd19B, aqz, mtCollI [Mitochondrial_membrane], CAHbeta, DmelCG8508, DmelCG4815, wupA
28.0-30.0 kDa	413	0	3	Actin_tubulin; Apoptosis; GAPDH_metabolic_controls; Histones; Mitochondrial_membrane; Notch; Protein_phosphatases; Ribosomal_proteins; Toll_lmd_NFKB; Ubiquitin_proteasome; Wnt_Wingless	high	Obp83cd, DmelCG12861, CG6171, 15010468, Obp83ef, Jon65Aiv, EG:BACR2583.6, lectin-29Ca, BaraA1, BaraA2, DmelCG32391, DmelCG34442, Obp56c, Jhbp7, Ag5r, CG11342, DmelCG9150, 312
40.0-42.0 kDa	316	2	0	Actin_tubulin; GAPDH_metabolic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG11367, DmelCG7530, DmelCG4983, DmelCG14445, CG5135, DmelCG2082, Cpr97Ea, Ubx, DmelCG17287, SP42, Uro, Bace, Gpdh2, Rtp6, Prp38, DmelCG5418, FMRFa, DmelCG8870
50.0-52.0 kDa	291	1	2	Actin_tubulin; Apoptosis; Autophagy; BMP_TGFbeta; Histones; Mitochondrial_membrane; Notch; Protein_kinases; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG33116, T3dh, DmelCG12061, DmelCG14718, Sry-delta [Notch], Uba3, acs, amls, Stard7, DmelCG14696, CG12258, DmelCG8563, Herp, Dbct, lP3K1, loqs, eIF4H2, nvd
20.0-22.0 kDa	441	0	2	Actin_tubulin; Apoptosis; Heat_shock_chaperones; Histones; Mitochondrial_membrane; RAS_ERK_MAPK; Ribosomal_proteins; Toll_lmd_NFKB; Ubiquitin_proteasome	high	retinin, Arf6, DmelCG34327, DCTN6-p27, mRpS10, DmelCG8300, Glo1, DmelCG34251, BcDNA:SD01241, dnd, mtND6, CheA7a, Stp24F, arx, Hand, lectin-33A, Arpc3B [Actin_tubulin], DmelCG33654
24.0-26.0 kDa	414	0	2	Apoptosis; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Notch; Protein_kinases; Ribosomal_proteins; Ubiquitin_proteasome	high	tun, DmelCG12535, DmelCG14932, DmelCG14086, DmelCG43312, Glaz, LSm-4, Polr3G, DmelCG33262, DmelCG14362, Mabi, CREG, DmelCG4000, DmelCG13226, DmelCG13002, Clap97D, DmelCG9782, Ccp84Ac
14.0-16.0 kDa	361	1	1	Apoptosis; Autophagy; Histones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_proteasome	high	DmelCG13050, CG13515, Cpr65Ea, BcDNA:RH04491, DmelCG13965, strat, dHYPK, DIGF-1, DmelCG44088, pal, BcDNA:AT27312, DmelCG15404, lTP, Obp18a, Cnep1r1, DmelCG13510, DmelCG31415, DmelCG34185
26.0-28.0 kDa	381	0	2	Actin_tubulin; GAPDH_metabolic_controls; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	high	CG11134, Gas41, Lip2, alphaTry, CG9154, DmelCG3699, DmelCG13616, DmelCG12729 [Ubiquitin_proteasome], BcDNA:GH06302, DmelCG13603, Vha26, Tsp42Eh, DmelCG33340, Osi9, CK02422a, DmelCG11672, Sld5, cerv
54.0-56.0 kDa	247	1	2	Apoptosis; GAPDH_metabolic_controls; Heat_shock_chaperones; Histones; Insulin_P13K_AKT_TOR; Mitochondrial_membrane; Protein_kinases; Ribosomal_proteins; Ubiquitin_proteasome; Wnt_Wingless	high	CapaR, DmelCG32204, dmGlut, BcDNA:RE64043, ana, sut2, DmelCG17782, Mur29B, Rh50, DmelCG11774, DmelCG11927, DmelCG5653, DmelCG14015, DmelCG18493, Dh44-R2, DmelCG15572, trbl, PIG-M
46.0-48.0 kDa	307	0	1	Actin_tubulin; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ribosomal_proteins; Toll_lmd_NFKB; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG1105, DmelCG14834, SptL, DmelCG15056, Or94a, CG12822-PA, DmelCG3987, tin, Or42b, Gr85a, Or69a, Egfrap, DmelCG7264, Yp3, btp1, DmelCG9372, DmelCG9775, Gr22a
64.0-66.0 kDa	224	1	1	Autophagy; BMP_TGFbeta; GAPDH_metabolic_controls; Heat_shock_chaperones; Histones; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ribosomal_proteins; Ubiquitin_proteasome	high	Atac3, nAChRalpha1, DmelCG11380, DmelCG17636, DmelCG11498, Alat, CG13007, Ttc26, Mfe2, DmelCG3544, kairos, d3635, CG33124-PB, Tdp1, CG31668-PB, gas, CG15332, Adck5
12.0-14.0 kDa	389	0	0	Actin_tubulin; Apoptosis; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Ribosomal_proteins	high	DmelCG42568, DmelCG14312, DmelCG14096, DmelCG44362, DmelCG42583, ATPsynGL [Mitochondrial_membrane], BaraC, Tst1, Cpr65Av, DmelCG13117, DmelCG32074, DmelCG15034, DmelCG32278, DmelCG43308, CG10674, DmelCG15200, DmelCG14984, ND-B12 [Mitochondrial_membrane]
42.0-44.0 kDa	313	0	2	Actin_tubulin; Apoptosis; JNK_MAPK; Notch; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	high	Rpn13 [Ubiquitin_proteasome], DmelCG7810, DmelCG17134, Fa2h, Pbp45, CG13886, Arp53D [Actin_tubulin], DmelCG18563, HLO3644, DmelCG17440, wkD, Ttc19, DmelCG14507, DmelCG18808, DmelCG12520, beg, Sdc, Rtc1
18.0-20.0 kDa	355	0	1	Actin_tubulin; Autophagy; Heat_shock_chaperones; Mitochondrial_membrane; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	high	ppl, mRpS17 [Ribosomal_proteins], EG-EG0003.8, DmelCG13994 [Ubiquitin_proteasome], DmelCG13560, BcDNA:AT27976, CCAP, DmelCG5500, DmelCG34312, Rpl34a [Ribosomal_proteins], DmelCG15368, Arcease, DmelCG12426, CG30182, DmelCG42655, Gadd45, DmelCG15023, RpS11 [Ribosomal_proteins]
58.0-60.0 kDa	306	0	1	Actin_tubulin; Apoptosis; Heat_shock_chaperones; Histones; Insulin_P13K_AKT_TOR; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG7910, Cyp6a9, CG40146, DmelCG14868, DmelCG7115, TKR86C, swi2, fon, CG11945, Klc, tup, Tpst, CCT2, coro, Dus1, DmelCG14655, Adgf-D, TBPH
70.0-72.0 kDa	180	2	2	Heat_shock_chaperones; Histones; Protein_kinases	high	(I2)k01209, Ir94a, Klp59C, mtDNA-helicase, eIF2A, DmelCG13786, DmelCG9521, DmelCG7246, eyg, Hayan, DmelCG4669, CCDC151, mxt, RSG7, Ance-4, Hsp70Aa [Heat_shock_chaperones], Hsp70Ab [Heat_shock_chaperones], dpr15
44.0-46.0 kDa	330	0	0	Actin_tubulin; Autophagy; Heat_shock_chaperones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ribosomal_proteins; Ubiquitin_proteasome; Wnt_Wingless	high	CG12294, CG33047, DmelCG11099, DmelCG43711, Gr10b, sug, DmelCG8539, CtsK2, a6, Pect, BcDNA:SD02419, eIF3m, DmelCG2267, Lip1t1, betaggt-l, Pk34A [Protein_kinases], anon-WO0118547.731, Dus4
52.0-54.0 kDa	265	0	3	Actin_tubulin; BMP_TGFbeta; Hedgehog; JNK_MAPK; Mitochondrial_membrane; Protein_kinases; Ubiquitin_proteasome; Wnt_Wingless	high	AcpH-1, ppk30, Jheh2, DmelCG8366, Drep4, FBgn, DmelCG42826, polyph, DmelCG7239, BcDNA:LP01340, ap, Ets65A, path, Sms, Dph2, Orc5, DmelCG8785, Argk3
38.0-40.0 kDa	339	0	1	Histones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG10657, BVRA, DmelCG12224, sro, AlkB, DmelCG11674, QC, CG18512, CG7840, Tssk2, DmelCG8746, UBL3 [Ubiquitin_proteasome], DmelCG14074, DmelCG3397, Rpn8 [Ubiquitin_proteasome], Cyt-c1L, DmelCG10931, mtTFB1
32.0-34.0 kDa	320	0	1	Actin_tubulin; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome; Wnt_Wingless	high	Osi3, Mtap, DmelCG11095, Cchl, DmelCG13622, SPH199, anon2A5 [Mitochondrial_membrane], DmelCG10587, DmelCG14329, DmelCG34295, CG17837, Nubpl, DmelCG34206, Ing5, DmelCG8252, DmelCG34011, Osi6, HDC06756
56.0-58.0 kDa	277	0	1	Apoptosis; Autophagy; GAPDH_metabolic_controls; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; Toll_lmd_NFKB; Ubiquitin_proteasome	elevated	Rpn3 [Ubiquitin_proteasome], gene, DmelCG16721, PIG-Q, Es2, Ofut2, emei, CtIP, DmelCG3610, Synd, PGRP-LC, DmelCG7352, SmydA-9, mthi6, B-H1, Cp7Fb, VGlut2, Vml
80.0-82.0 kDa	128	1	3	Apoptosis; BMP_TGFbeta; Heat_shock_chaperones; JNK_MAPK; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome; Wnt_Wingless	elevated	DmelCG33337, Exo70, Mks1, qtc, Cka, Nep14, DmelCG12239, sens-2, Trim9 [Ubiquitin_proteasome], DmelCG6225, DmelCG13671, m-cup, PIG-Z, Soc36E, Pink1 [Protein_kinases], gro, DmelCG12010, Mipep
36.0-38.0 kDa	329	0	0	Apoptosis; Autophagy; Notch; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	elevated	EG:BACR43E12.6, Sbp2, PPP1R15 [Protein_phosphatases], Tengl1, DmelCG14511, AnxB9, Mdh1, DmelCG1275, jet, Sr-CIII, wmd, anon-WO0242455.53, Zip48C, TBC167, JMJD7, elgi [Ubiquitin_proteasome], Ski8, Gmer
10.0-12.0 kDa	343	0	0	Apoptosis; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_proteasome	elevated	pre-mod(mdg4)-AA, SLIRP1, DmelCG7377, CG9167, DmelCG43405, hemo, BcDNA:RE60135, Tim8 [Mitochondrial_membrane], DmelCG13069, VhaM9-7-b, DmelCG33268, inaF-A, DmelCG43982, DmelCG42806, UQCRC-Q, DmelCG45486, DmelCG42581, Triap1 [Apoptosis]
68.0-70.0 kDa	195	1	1	Autophagy; Heat_shock_chaperones; Hedgehog; Notch; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome	elevated	lovit, CG3430, Cnx99A, DmelCG8547, Dus3, Tkt, anon-WO0118547.91, Menl-1, myo, Pgant8, Mettl3, dind, ppk9, DmelCG17803, Moe, sav, TyrR, DmelCG33296
60.0-62.0 kDa	241	0	2	Actin_tubulin; GAPDH_metabolic_controls; Heat_shock_chaperones; Histones; Notch; Protein_kinases; Toll_lmd_NFKB; Ubiquitin_proteasome	elevated	Kap-alpha1, gish [Protein_kinases], C901 [Notch], Ubi-p5E [Ubiquitin_proteasome], CG17099, LPCAT, Cyp9h1, Dgkepsilon, DmelCG14223, DmelCG9784, Eogt, Ac, Salt, klg, Ugt49B1, CG13382, DmelCG3347, DmelCG32454

Common loading/compartiment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
14.39	Atg8a	Q9W2S2	Q9W2S2_DROME	Autophagy-related 8a, isoform A	14.0-16.0 kDa
16.02	RpL32	P04359	RL32_DROME	Large ribosomal subunit protein eL32	16.0-18.0 kDa
23.33	Rab7	O76742	RAB7_DROME	Ras-related protein Rab7	22.0-24.0 kDa
23.96	Rab5	Q9V3I2	RAB5_DROME	Ras-related protein Rab5	22.0-24.0 kDa
30.55	porin	Q94920	VDAC_DROME	Voltage-dependent anion-selective channel	30.0-32.0 kDa
35.35	Gapdh1	P07486	G3P1_DROME	Glyceraldehyde-3-phosphate dehydrogenase 1	34.0-36.0 kDa
35.37	Gapdh2	P07487	G3P2_DROME	Glyceraldehyde-3-phosphate dehydrogenase 2	34.0-36.0 kDa
41.82	Act5C	P10987	ACT1_DROME	Actin-5C	40.0-42.0 kDa
41.82	Act42A	P02572	ACT2_DROME	Actin-42A	40.0-42.0 kDa
49.89	alphaTub84D	P06605	TBA3_DROME	Tubulin alpha-3 chain	48.0-50.0 kDa
49.91	alphaTub84B	P06603	TBA1_DROME	Tubulin alpha-1 chain	48.0-50.0 kDa
50.15	betaTub56D	Q24560	TBB1_DROME	Tubulin beta-1 chain	50.0-52.0 kDa
54.11	ATPsynbeta	Q05825	ATPB_DROME	ATP synthase subunit beta, mitochondrial	54.0-56.0 kDa
65.30	ref(2)P	P14199	REF2P_DROME	Protein ref(2)P	64.0-66.0 kDa
68.02	Cnx99A	Q9VAL7	Q9VAL7_DROME	Calnexin 99A, isoform A	68.0-70.0 kDa
71.13	Hsc70-4	P11147	HSP7D_DROME	Heat shock 70 kDa protein cognate 4	70.0-72.0 kDa
71.30	Lam	P08928	LMNB_DROME	Lamin	70.0-72.0 kDa
81.87	Hsp83	P02828	HSP83_DROME	Heat shock protein 83	80.0-82.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
0.0-2.0 kDa	7	0	0		routine	tal-1A, tal-2A, tal-3A, DmelCG43739, CG14615-utr, CG6424-5utr, DmelCG46270
2.0-4.0 kDa	49	0	0	Ribosomal_proteins	routine	CG14253-5utr, DmelCG43178, DmelCG46525, DmelCG45088, MSAmiP, DmelCG43200, noncoding_5744, MIP3715, ORF_1, Acq98AB, MIP3715, CG13627-5utr, DmelCG46335, CG43200-3utr, Mst40, DmelCG46310, DmelCG45307, CR33939, CG7708-5utr
4.0-6.0 kDa	104	0	0	Ribosomal_proteins	routine	DmelCG43965, chr2L_11663230_11663806, BomS3, DmelCG45099, DmelCG43814, Mst57Db, BomS6, gdl-ORF39, DmelCG43251, BomS5, lincRNA.174, DmelCG43645, Osl4, Hdac3-5utr, DmelCG43390, Dso1, MtnB, DmelCG43694
6.0-8.0 kDa	189	0	0	Mitochondrial_membrane; Ribosomal_proteins	elevated	Tom7, DmelCG43610, DmelCG46320, DmelCG43618, DmelCG46456, CG4959, Ang, DmelCG43230, DmelCG44040, BcDNA-RH07028, DmelCG45089, CR43920, DmelCG43185, ND-MNLL [Mitochondrial_membrane], Dup998, DmelCG46429, DmelCG44355, UQCR-6.4
8.0-10.0 kDa	290	0	0	Heat_shock_chaperones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_proteasome	elevated	lawc, UQCR-6.4L, E[sp]m6-BFM, p8, BomT2, DmelCG17776 [Mitochondrial_membrane], DmelCG43182, Ggamma1, DmelCG11458, DmelCG46433, DmelCG44142, DrsI6, NEST-bs34b09, CG15281, ksh, DmelCG43169, DmelCG42649, Rpl38 [Ribosomal_proteins]
10.0-12.0 kDa	343	0	0	Apoptosis; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_proteasome	elevated	pre-mod(mdg4)-AA, SLIRP1, DmelCG7377, CG9167, DmelCG43405, hemo, BcDNA:RE60135, TimB [Mitochondrial_membrane], DmelCG13069, VhaM9.7-b, DmelCG33268, inaF-A, DmelCG43982, DmelCG42806, UQCR-Q, DmelCG45486, DmelCG42581, Triap1 [Apoptosis]
12.0-14.0 kDa	389	0	0	Actin_tubulin; Apoptosis; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Ribosomal_proteins	high	DmelCG42568, DmelCG14312, DmelCG14096, DmelCG44362, DmelCG42583, ATPsynGL [Mitochondrial_membrane], BaraC, Tst1, Cpr65Av, DmelCG13117, DmelCG32074, DmelCG15034, DmelCG32278, DmelCG43308, CG10674, DmelCG15200, DmelCG14984, ND-B12 [Mitochondrial_membrane]
14.0-16.0 kDa	361	1	1	Apoptosis; Autophagy; Histones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_proteasome	high	DmelCG13050, CG13515, Cpr65Ea, BcDNA-RH04491, DmelCG13965, strat, dHYPK, DIGF-1, DmelCG44088, pal, BcDNA:AT72312, DmelCG15404, ITP, Obp18a, Cnep1r1, DmelCG13510, DmelCG31415, DmelCG34185
16.0-18.0 kDa	424	1	3	Actin_tubulin; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Notch; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	high	Chrac-16, DmelCG13689, DmelCG13239, DmelCG7781, Rbp1, Rps23 [Ribosomal_proteins], Rpl32 [Ribosomal_proteins], Npc2a, ND-B14.5AL [Mitochondrial_membrane], Rpl28 [Ribosomal_proteins], DmelCG9568, DmelCG11663, Pbus, DmelCG11300, MED21, DmelCG14658, Blos3, BcDNA:RE30174
18.0-20.0 kDa	355	0	1	Actin_tubulin; Autophagy; Heat_shock_chaperones; Mitochondrial_membrane; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	high	ppl, mRps17 [Ribosomal_proteins], EG-EG0003.8, DmelCG13994 [Ubiquitin_proteasome], DmelCG13560, BcDNA:AT72976, CCAP, DmelCG5500, DmelCG34312, Rpl34a [Ribosomal_proteins], DmelCG15368, Arcease, DmelCG12426, CG30182, DmelCG42655, Gadd45, DmelCG15023, RpS11 [Ribosomal_proteins]
20.0-22.0 kDa	441	0	2	Actin_tubulin; Apoptosis; Heat_shock_chaperones; Histones; Mitochondrial_membrane; RAS_ERK_MAPK; Ribosomal_proteins; Toll_lmd_NFKB; Ubiquitin_proteasome	high	retinin, Arf6, DmelCG34327, DCTN6-p27, mRps10, DmelCG8300, Glo1, DmelCG34251, BcDNA:SD01241, dnd, mtND6, CheA7a, Slp24F, arx, Hand, lectin-33A, Arpc3B [Actin_tubulin], DmelCG33654
22.0-24.0 kDa	358	2	2	Actin_tubulin; Heat_shock_chaperones; Mitochondrial_membrane; RAS_ERK_MAPK; Ribosomal_proteins; Ubiquitin_proteasome	high	Clb2, DmelCG31109, DmelCG11226, CG14286, DmelCG14495, ND-SGDH [Mitochondrial_membrane], BcDNA:GH111730, DmelCG16813, VhaPAP1-1, CG15250, DmelCG42830, DmelCG14455, BP1005, DmelCG32797, Rab18, DmelCG14777, DmelCG34138, DmelCG3955
24.0-26.0 kDa	414	0	2	Apoptosis; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Notch; Protein_kinases; Ribosomal_proteins; Ubiquitin_proteasome	high	tun, DmelCG12535, DmelCG14932, DmelCG14086, DmelCG43312, GLaz, LSm-4, Polr3G, DmelCG33262, DmelCG14362, Mabi, CREG, DmelCG4000, DmelCG13226, DmelCG13002, Clap97D, DmelCG782, Ccp84Ac
26.0-28.0 kDa	381	0	2	Actin_tubulin; GAPDH_metabolic_controls; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	high	CG11134, Gas41, Lipt2, alphaTry, CG9154, DmelCG3699, DmelCG13616, DmelCG12729 [Ubiquitin_proteasome], BcDNA:GH06302, DmelCG13603, Vha26, Tsp42Eh, DmelCG33340, Osi9, CK02422a, DmelCG11672, Sld5, cerv
28.0-30.0 kDa	413	0	3	Actin_tubulin; Apoptosis; GAPDH_metabolic_controls; Histones; Mitochondrial_membrane; Notch; Protein_phosphatases; Ribosomal_proteins; Toll_lmd_NFKB; Ubiquitin_proteasome; Wnt_Wingless	high	Obp83cd, DmelCG12861, CG6171, 15010468, Obp83ef, Jon65Aiv, EG:BACR25B3.6, lectin-29Ca, BaraA1, BaraA2, DmelCG32391, DmelCG34442, Obp56c, Jhnp7, Ag5r, CG11342, DmelCG9150, 312
30.0-32.0 kDa	362	1	1	Actin_tubulin; Apoptosis; Autophagy; GAPDH_metabolic_controls; Histones; JNK_MAPK; Mitochondrial_membrane; Notch; Protein_phosphatases; Ribosomal_proteins; Toll_lmd_NFKB; Ubiquitin_proteasome	high	Fire-like, DmelCG10053, DmelCG13408, Jhnp5, BcDNA:SD21514, DmelCG13996, lp259, Fst, Ric, Osi7, Q9VXA9, fd19B, aqz, mt.CollII [Mitochondrial_membrane], CAHbeta, DmelCG8508, DmelCG4815, wupA
32.0-34.0 kDa	320	0	1	Actin_tubulin; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome; Wnt_Wingless	high	Osi3, Mtap, DmelCG11095, Cchl, DmelCG13622, SPH199, anon2A5 [Mitochondrial_membrane], DmelCG10587, DmelCG14329, DmelCG34295, CG17837, Nutpl, DmelCG34206, lng5, DmelCG8252, DmelCG34011, Osi6, HDC06756
34.0-36.0 kDa	352	2	3	Actin_tubulin; Apoptosis; GAPDH_metabolic_controls; Notch; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome; Wnt_Wingless	high	TORIP, DmelCG10973, Teh1, DmelCG11617, RplL5 [Ribosomal_proteins], DmelCG34444, CAH14, LPP-like, DmelCG6067, DmelCG10465, MET30, DmelCG46434, DmelCG6972, DmelCG32371, Surf1, Jhamt, DmelCG14420, Osi18
36.0-38.0 kDa	329	0	0	Apoptosis; Autophagy; Notch; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_proteasome	elevated	EG:BACR43E12.6, Sbp2, PPP1R15 [Protein_phosphatases], Teng1, DmelCG14511, Anx89, Mdh1, DmelCG1275, jett, Sr-ClII, wmd, anon-WO0242455.53, Zip48C, TBC1d7, JMDJ7, elgi [Ubiquitin_proteasome], Ski8, Gmer
38.0-40.0 kDa	339	0	1	Histones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG10657, BVRA, DmelCG12224, sro, AlkB, DmelCG11674, QC, CG18512, CG7840, Tssk2, DmelCG8746, UBL3 [Ubiquitin_proteasome], DmelCG14074, DmelCG3397, Rpn8 [Ubiquitin_proteasome], Cyt-c1L, DmelCG10931, mtTFB1
40.0-42.0 kDa	316	2	0	Actin_tubulin; GAPDH_metabolic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG11367, DmelCG7530, DmelCG4983, DmelCG14445, CG5135, DmelCG2082, Cpr9Ea, Ubx, DmelCG17287, SP42, Uro, Bace, Gpdh2, Rbp6, Prp38, DmelCG5418, FMRFa, DmelCG8870
42.0-44.0 kDa	313	0	2	Actin_tubulin; Apoptosis; JNK_MAPK; Notch; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ubiquitin_proteasome	high	Rpn13 [Ubiquitin_proteasome], DmelCG7810, DmelCG17134, Fa2h, Pbp45, CG13886, Arp5D [Actin_tubulin], DmelCG18563, HL03644, DmelCG17440, wkdi, Ttc19, DmelCG14507, DmelCG18808, DmelCG12520, beg, Sdc, Rtc1
44.0-46.0 kDa	330	0	0	Actin_tubulin; Autophagy; Heat_shock_chaperones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ribosomal_proteins; Ubiquitin_proteasome; Wnt_Wingless	high	CG12294, CG33047, DmelCG11099, DmelCG43711, Gr10b, sug, DmelCG8539, CtskK2, a6, Pect, BcDNA:SD02419, eIF3m, DmelCG2267, Lipt1, betaggt-I, Pk34A [Protein_kinases], anon-WO0118547.731, Dus4
46.0-48.0 kDa	307	0	1	Actin_tubulin; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ribosomal_proteins; Toll_lmd_NFKB; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG1105, DmelCG14834, SppL, DmelCG15056, Or94a, CG12822-PA, DmelCG3987, tin, Or42b, Gr85a, Or69a, Egfrap, DmelCG7264, Yp3, bip1, DmelCG9372, DmelCG9775, Gr22a
48.0-50.0 kDa	300	2	0	Actin_tubulin; Apoptosis; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ubiquitin_proteasome; Wnt_Wingless	high	GalT5, pre-Iola-G, [K3]neo38, anon-WO0118547.264, PolD2 [Notch], sut4, rho-4, DmelCG10376, Diap1 [Apoptosis], CG13770, DmelCG3635, DmelCG10559, carboxypeptidase, 6959, Eig71Ee, cpt, M1BP, brwl
50.0-52.0 kDa	291	1	2	Actin_tubulin; Apoptosis; Autophagy; BMP_TGFbeta; Histones; Mitochondrial_membrane; Notch; Protein_kinases; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG33116, T3dh, DmelCG12061, DmelCG14718, Sry-delta [Notch], Uba3, acs, amls, Stard7, DmelCG14696, CG12258, DmelCG8563, Herp, Dbct, IPK31, loqs, eIF4H2, nvd
52.0-54.0 kDa	265	0	3	Actin_tubulin; BMP_TGFbeta; Hedgehog; JNK_MAPK; Mitochondrial_membrane; Protein_kinases; Ubiquitin_proteasome; Wnt_Wingless	high	Acph-1, ppk30, Jheh2, DmelCG8366, Drep4, FBgn, DmelCG42826, polyph, DmelCG7239, BcDNA:LP01340, ap, EisB5A, path, Sms, Dph2, Orc5, DmelCG8785, Argk3
54.0-56.0 kDa	247	1	2	Apoptosis; GAPDH_metabolic_controls; Heat_shock_chaperones; Histones; Insulin_Pi3K_AKT_TOR; Mitochondrial_membrane; Protein_kinases; Ribosomal_proteins; Ubiquitin_proteasome; Wnt_Wingless	high	CapaR, DmelCG32204, dmGlut, BcDNA:RE64043, ana, sut2, DmelCG17782, Mur29B, Rh50, DmelCG1774, DmelCG11927, DmelCG5653, DmelCG14015, DmelCG18493, Dh44-R2, DmelCG15572, trbi, PIG-M
56.0-58.0 kDa	277	0	1	Apoptosis; Autophagy; GAPDH_metabolic_controls; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; Toll_lmd_NFKB; Ubiquitin_proteasome	elevated	Rpn3 [Ubiquitin_proteasome], gene, DmelCG16721, PIG-Q, Es2, Ofut2, ernei, ClIP, DmelCG3610, Synd, PGRP-LC, DmelCG7352, SmydA-9, mthl6, B-H1, Cp7Fb, Vglut2, Vml
58.0-60.0 kDa	306	0	1	Actin_tubulin; Apoptosis; Heat_shock_chaperones; Histones; Insulin_Pi3K_AKT_TOR; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome; Wnt_Wingless	high	DmelCG7910, Cyp6a9, CG40146, DmelCG14868, DmelCG7115, TkR86C, swi2, fon, CG11945, Klc, tup, Tpst, CCT2, coro, Dus1, DmelCG14655, Adgf-D, TBPH
60.0-62.0 kDa	241	0	2	Actin_tubulin; GAPDH_metabolic_controls; Heat_shock_chaperones; Histones; Notch; Protein_kinases; Toll_lmd_NFKB; Ubiquitin_proteasome	elevated	Kap-alpha1, gish [Protein_kinases], C901 [Notch], Ubi-p5E [Ubiquitin_proteasome], CG17099, LPCAT, Cyp9h1, Dgkpsilon, DmelCG14223, DmelCG9784, Eogt, AC, Saft, klg, Ugt49B1, CG13382, DmelCG3347, DmelCG32454
62.0-64.0 kDa	207	0	0	Histones; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome	elevated	RluA-2, DmelCG5191, NimA, DmelCG42304, Ada2b, DmelCG7492, emp, anon-WO0140519.45, DmelCG1271, BcDNA:GH24648, swa, DmelCG3281, DmelCG6479, mthl11, Jhedup, Ptp61F [Protein_phosphatases], Snnp1, PH4alphaNE2
64.0-66.0 kDa	224	1	1	Autophagy; BMP_TGFbeta; GAPDH_metabolic_controls; Heat_shock_chaperones; Histones; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ribosomal_proteins; Ubiquitin_proteasome	high	Atac3, nAChRalphal1, DmelCG11380, DmelCG17636, DmelCG11498, Alat, CG13007, Ttc26, Mfe2, DmelCG3544, kairos, d3635, CG33124-PB, Tdp1, CG31668-PB, gas, CG15332, Adck5
66.0-68.0 kDa	183	0	1	Actin_tubulin; BMP_TGFbeta; Insulin_Pi3K_AKT_TOR; Mitochondrial_membrane; Notch; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome; Wnt_Wingless	elevated	CG10428, CG10532 [Protein_kinases], DmelCG13870, DS06874.7, Slo45-1, RanGAP, pie, DmelCG11964 [Wnt_Wingless], DmelCG10801, Ugt305A1, dyl, vlc, ChLD3, E(var)3-9, DmelCG4382, schuy, Mst84B, opa
68.0-70.0 kDa	195	1	1	Autophagy; Heat_shock_chaperones; Hedgehog; Notch; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome	elevated	lovit, CG3430, Cnx99A, DmelCG8547, Dus3, Tkt, anon-WO0118547.91, Menl-1, myo, Pgan8, Mett13, dind, ppk9, DmelCG17803, Moe, sav, TyrR, DmelCG32396
70.0-72.0 kDa	180	2	2	Heat_shock_chaperones; Histones; Protein_kinases	high	[J2]k01209, Ir94a, Klp59C, mtDNA-helicase, eIF2A, DmelCG13786, DmelCG9521, DmelCG7246, eyg, Hayan, DmelCG4669, CCDC151, mxt, RSG7, Ance-4, Hsp70Aa [Heat_shock_chaperones], Hsp70Ab [Heat_shock_chaperones], dpr15

MW bin	N	Markers	Signal	Families	Priority	Examples / audit familles
72.0-74.0 kDa	147	0	0	Actin_tubulin; Heat_shock_chaperones; Protein_kinases; Protein_phosphatases; Wnt_Wingless	elevated	Alg9, pasha, DmelCG4459, Tdc2, DmelCG13605, Fim, dah [Actin_tubulin], Pgan5, byn, CYLD, Rph, Dmtn, IntS10, Hsc70-3 [Heat_shock_chaperones], PK2-R1, Gat, Idua, DmelCG13458
74.0-76.0 kDa	144	0	2	Actin_tubulin; Autophagy; BMP_TGFbeta; Heat_shock_chaperones; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Toll_lmd_NFkB; Ubiquitin_proteasome	elevated	DmelCG5953, RspH3, Pqant4, Myb, Pappss, dsf, Hsc70-5 [Heat_shock_chaperones], DmelCG9289, DmelCG8925, Cip4, DmelCG5687, Adar, HBS1, Klf3C, Sh, Pltc3, Cp110, Lasp
76.0-78.0 kDa	160	0	1	Autophagy; BMP_TGFbeta; Heat_shock_chaperones; Histones; Insulin_Pi3K_AKT_TOR; Mitochondrial_membrane; Notch; Protein_kinases; Ribosomal_proteins	elevated	CG2614, Acox57D-p, pira, DmelCG11369, DmelCG12990, Fmr1, DmelCG13097, cg3225, DmelCG4822, Gyc98Da, pon, Pak [Protein_kinases], CarT, DmelCG10481, Klp64D, sbr, DmelCG31693, IP3K2
78.0-80.0 kDa	121	0	0	Protein_kinases; Ubiquitin_proteasome; Wnt_Wingless	elevated	DmelCG5608, Gfat1, DmelCG15473, Pep, DmelCG13325, CG9561, gye, IME2 [Protein_kinases], ZnT33D, Ir75d, DmelCG1719, DmelCG4712, Ror [Protein_kinases], CG9213, nvy, DmelCG4576, feo, Nopp140
80.0-82.0 kDa	128	1	3	Apoptosis; BMP_TGFbeta; Heat_shock_chaperones; JNK_MAPK; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome; Wnt_Wingless	elevated	DmelCG33337, Exo70, Mks1, qtc, Cka, Nep114, DmelCG12239, sens-2, Trim9 [Ubiquitin_proteasome], DmelCG6225, DmelCG13671, m-cup, PIG-Z, Socs36E, Pimk1 [Protein_kinases], gro, DmelCG12010, Mipep
82.0-84.0 kDa	114	0	1	GAPDH_metabolic_controls; Protein_kinases; RAS_ERK_MAPK; Ubiquitin_proteasome	elevated	Mitofilin, Sec10, BTBD9, DmelCG5664, Nep7, RspH4a, FoxK, CG18499, dx, CG11074, DmelCG34141, ths, ncon, DmelCG44435, yin, Orc3, Mcm5, REPTOR
84.0-86.0 kDa	92	0	2	Actin_tubulin; Notch; Protein_kinases; Ubiquitin_proteasome	elevated	t-GripB4 [Actin_tubulin], Pbp95, CG42342, Mtpalpha, Cog7, Ir40a, P5CS [Notch], bnl, l2jdfl, Nsun2, CysRS, crn, EG-BACR421l7.8, CG5100, anon-WO0140519.42, TTL6B [Actin_tubulin], DmelCG30471, Fpgs2
86.0-88.0 kDa	89	0	1	Actin_tubulin; Histones; JAK_STAT; Protein_kinases	elevated	TTL3B [Actin_tubulin], Crtp, Tmem63, Best1, DmelCG7248, Cad96Ca, Oseg5, tw, Vrp1, Unc-115b, anon-WO0172774.162, DmelCG53355, DmelCG16786, DmelCG10663, BcDNA:AT10318, IKKbeta, DmelCG9467, Stat92E [JAK_STAT]
88.0-90.0 kDa	86	0	2	GAPDH_metabolic_controls; Histones; Notch; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome	elevated	Xpd, DmelCG13907, Fili, SF3a1, Efhc1.2, mal, Nep2, CG10958, trbd [Ubiquitin_proteasome], Sobp, Klp68D, BcDNA:HL07910, su(f), bru1, aru, DmelCG31784, zf30C, Oatp33EB
90.0-92.0 kDa	80	0	1	Heat_shock_chaperones; Histones; Protein_kinases; Toll_lmd_NFkB; Ubiquitin_proteasome; Wnt_Wingless	elevated	DmelCG17111, CT30663, Spg7, sca, CG31530, klu, DmelCG32320, Gp93 [Heat_shock_chaperones], Tehao, fu [Protein_kinases], hay, CG12084, roq [Ubiquitin_proteasome], Muc96D, CG5033, Mayo, Dyb, Edem2
92.0-94.0 kDa	76	0	0	Autophagy; Histones; Protein_kinases; Ubiquitin_proteasome	elevated	DmelCG13643, DmelCG12241, DmelCG8613, POSH, p115, DmelCG17732, Usp5 [Ubiquitin_proteasome], Gcn5 [Histones], DmelCG5290, topi, Slm, hrg, DmelCG15160, br, Tsf2, Klp67A, Mcm6, Not3
94.0-96.0 kDa	76	0	1	Actin_tubulin; Protein_kinases; Ubiquitin_proteasome; Wnt_Wingless	elevated	Hdc, SrpK79D [Protein_kinases], Pdk1 [Protein_kinases], kek6, Brd8, CG12167, CG42506, DmelCG31195, EG-125H10.1, prage, CuI4, eEF2, Hll, Noc3, GABA-B-R1, kek1, DmelCG10333, DmelCG32668 [Wnt_Wingless]
96.0-98.0 kDa	77	0	1	Autophagy; Hedgehog; Protein_kinases	elevated	Zn72D, lmd, CG17740, DmelCG11248, Targ, mus304, Lap1, GCS1, D19A, Atg9 [Autophagy], tara, Dnaaf5, lin, nan, Snrk [Protein_kinases], DmelCG17816, Reps, 148274_at
98.0-100.0 kDa	71	0	0	Actin_tubulin; Histones; Insulin_Pi3K_AKT_TOR	routine	CIC-c, DmelCG31327, simj, CG18546, CG9416-PA, in, DmelCG14085, DmelCG2938, Ance-3, DmelCG10062, rgn, DmelCG17739, DmelCG10081, Hmgcr, DmelCG13160, Appl, PF1, Sdr [Insulin_Pi3K_AKT_TOR]
100.0-102.0 kDa	73	0	1	Actin_tubulin; Protein_kinases; Ubiquitin_proteasome	elevated	mip120, sano, goe, Su(z)12, DmelCG30043, Lerp, Vps39, Nep5, DmelCG45067, DmelCG5787, mfas, pad, Pcf1f, DmelCG8665, NT1, DmelCG13008, Mcm2, mahe
102.0-104.0 kDa	74	0	0	Actin_tubulin; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome	elevated	dyst, CG15175, Pde8, Saf-B, CT7856, DmelCG1737, GluRIIE, DmelCG6834, rgr, Rpn1 [Ubiquitin_proteasome], Prm, alpha-Cat, CG2164, kek5, DmelCG9861, gem, DmelCG14183, bi
104.0-106.0 kDa	71	0	4	Actin_tubulin; Histones; Insulin_Pi3K_AKT_TOR; Protein_kinases; Toll_lmd_NFkB	elevated	Toll-9 [Toll_lmd_NFkB], Spgs, dgo, DmelCG6059, Pkd2, CR31846, Atxn7, CG17740, DmelCG10413, Shark [Protein_kinases], GluRIIB, DmelCG9411, Cep104, DmelCG4907, Vha100-3, smid, TTL5 [Actin_tubulin], ukar
106.0-108.0 kDa	61	0	1	Insulin_Pi3K_AKT_TOR; Notch; Protein_phosphatases; Ubiquitin_proteasome	elevated	DmelCG2839, frtz, tmm, su(w[a]), Calk, DmelCG30484, Ir8a, Vinc, Pdpr, twy, crol, Actn, CG5253, Lztr1, DmelCG11951, ssp2, CG31177, side
108.0-110.0 kDa	76	0	1	Actin_tubulin; Protein_kinases; Protein_phosphatases; RAS_ERK_MAPK; Ubiquitin_proteasome	elevated	SP1029, Rhau, CenG1A, pepCG2111, melt, Ir25a, pHCl-1, EMSY, BBS9, Patr-1, CG33090, Orp8, DmelCG31343, tio, DmelCG14650, itf, cg9346, Mco1
110.0-112.0 kDa	56	0	1	Actin_tubulin; Protein_kinases; Ubiquitin_proteasome	elevated	Rgl, side-IV, Cse1, stau, wrw, DmelCG31077, brat, Khc, mip130, stg1, Cals, dbr, DmelCG14309, DmelCG4849, Rnb, RhoGAP54D, del, KCNQ
112.0-114.0 kDa	68	0	0	Actin_tubulin; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome; Wnt_Wingless	elevated	CG15516, egl, DmelCG17707, Dis3, stx, Cdc39, TTL3A [Actin_tubulin], defl, Nmdar1, low, E23, JYalpha, Rev1, Wnt5 [Wnt_Wingless], MstProx, MetRS, solo, DmelCG6686
114.0-116.0 kDa	49	0	0	Actin_tubulin; Notch; Protein_kinases; Protein_phosphatases; Ubiquitin_proteasome	routine	scny [Ubiquitin_proteasome], DmelCG32107, DmelCG4887, eater, DmelCG5877, DmelCG16952, Edem1, otk [Protein_kinases], Gcna, S1P, DmelCG14442, Ipo9, side-II, LManII, DmelCG12075, Pdzd8, DmelCG15635, Pcl
116.0-118.0 kDa	44	0	1	Hedgehog; Ubiquitin_proteasome	routine	CG7198, Clk, Usp1 [Ubiquitin_proteasome], CG8775, unc-5, Gem3, DmelCG16989, Ddr, smo [Hedgehog], Unr, Hcs, asl, zfh1, DmelCG13457, Apc2, mwh, SPoCk, ome
118.0-120.0 kDa	41	0	0	Protein_kinases; Ubiquitin_proteasome	routine	Kap3, DmelCG31897, Dis32, Tsp, CycT, Tao [Protein_kinases], lds, Ack, ebo, sfl, scx, CG7388, CngB, DmelCG7058, DmelCG43968, Glt, srl, ValRS
120.0-122.0 kDa	43	0	0	Hedgehog; Insulin_Pi3K_AKT_TOR; Protein_kinases; Ubiquitin_proteasome	routine	I(2)05287, Ire1 [Protein_kinases], Lrprrc2, IRSp53 [Insulin_Pi3K_AKT_TOR], DmelCG31586, lincRNA:S6902, Prp39, DmelCG17097, PNUTS, milk, CG14535, Ars, cg33288, DmelCG6280, Lpin, boi [Hedgehog], Gp150, Apl
122.0-124.0 kDa	29	0	0	Actin_tubulin; Protein_kinases; Ubiquitin_proteasome	routine	Impbeta11, DmelCG13251, DmelCG13921, Nmdar2, GlyT, DmelCG10588, wts [Protein_kinases], DmelCG3744, DmelCG8155, Cpr76Bd, LManII, Psa, emb, Ogdh1, Nin, Liprin-gamma, TTL4A [Actin_tubulin], dia
124.0-126.0 kDa	42	0	2	JNK_MAPK; Notch; Protein_kinases; RAS_ERK_MAPK; Toll_lmd_NFkB	elevated	MEP-1, DmelCG6614, dati, CG12982, DmelCG15599, Pde6, pico, SREBP, BEST-GM10035, ush, spd-2, Pacs, scb, thw, Dna2, FBX011, Ti [Toll_lmd_NFkB], iav
126.0-128.0 kDa	43	0	1	Actin_tubulin; Insulin_Pi3K_AKT_TOR; Protein_kinases; RAS_ERK_MAPK	elevated	side-V, pic, 2mit, par-1, eag, wry, Eip93F, DmelCG4467, mura, Map205, alpha-Man-IIa, subduced, I(2)gl, sac, I(3)l1231, P3K92E [Insulin_Pi3K_AKT_TOR], X11L, Trpgamma
128.0-130.0 kDa	43	0	0	Histones; Protein_kinases; Ubiquitin_proteasome	routine	CG2982, fw, mus312, DmelCG12983, DmelCG14441, Cdk12 [Protein_kinases], Hip1, Polr1B, Lk6 [Protein_kinases], Polr3B, GH25970, 1273, DmelCG43333, ACXE, DmelCG3356 [Ubiquitin_proteasome], stf, Flacc, Ankie2
130.0-132.0 kDa	36	0	1	Protein_kinases; Toll_lmd_NFkB; Ubiquitin_proteasome	routine	cdi, bon, SA1, GCC185, MBD-R2, DmelCG18596, Oatp30B, CG31149, chn, Ac3, Usp7 [Ubiquitin_proteasome], Ctr9, mmm, slam, Magl, CG12211, Uba1 [Ubiquitin_proteasome], HDAC6
132.0-134.0 kDa	32	0	1	GAPDH_metabolic_controls; Hedgehog; Protein_kinases; Wnt_Wingless	routine	Sym, pan [Wnt_Wingless], RasGAP1, DmelCG4835, PIP82, CG12967, hppy [Protein_kinases], Pcb, DmelCG1695, Nrd1, Dgk, slo, CG8970, Msh6, Sordl [GAPDH_metabolic_controls], cos, DmelCG7728, Frl
134.0-136.0 kDa	36	0	2	Actin_tubulin; Heat_shock_chaperones; JAK_STAT; Protein_kinases	elevated	DmelCG5756, Polr2B, kz, srp, L, BcDNA:GH07466, SMC2, Eps-15, hbs, ACXD, DmelCG18368, CG32809, Tbcd [Actin_tubulin; Heat_shock_chaperones], Synj, Blimp-1, LeuRS, Sgs1, Nef-A
136.0-138.0 kDa	36	0	0	Histones; JNK_MAPK; Protein_kinases; Protein_phosphatases	routine	scra, ATP7, DmelCG1647, Top3alpha, pdm3, HDACA [Histones], Prp5, Octbeta3R, Mat89Ba, kuz, DmelCG6230, GramD1B, SF3b3, syd [JNK_MAPK], CG12644, CG7065, anon-15Db, Nlg2
138.0-140.0 kDa	47	0	1	Actin_tubulin; Ubiquitin_proteasome	routine	CG32543, bun, tapas, Sec31, prom, GABA-B-R2, Ube4B [Ubiquitin_proteasome], AOX4, DmelCG3632, Fur1, PMCA, Ret, DmelCG42260, Usp15-31, Asator [Actin_tubulin], pds5, rdgB, PAN2
140.0-142.0 kDa	28	0	0	Histones; Notch	routine	SMC3, Upf2, DmelCG9727, AOX2, pod1, CT33170, CG9776, CG18490, Gfrl, CG7766, disco-r, DmelCG30122, EHR, mu2, Ca-Ma2l [Notch], pk, isha, jvl
142.0-144.0 kDa	40	0	2	Hedgehog; JAK_STAT	elevated	sl, DmelCG9068, DmelCG32719, dome [JAK_STAT], DmelCG6938, SCAP, trp, Mdr49, Fife, Ae2, Elp1, Ac76E, DmelCG4213, ptc [Hedgehog], a, SMC1, RhoGAP102A, Muc11A
144.0-146.0 kDa	23	0	0	Protein_phosphatases	routine	Xpc, Dp1, blo, kibra, RIGEF, Gart, robo3, gprs, DmelCG4393, inaE, CG13520, Ptp99A [Protein_phosphatases], DmelCG33125, Cap-D3, Dme_CG9270, Mdr50, Nrx-IV, pwn
146.0-148.0 kDa	37	0	0	BMP_TGFbeta; Histones	routine	cyst, Nhe2, DmelCG1582, row, CG13256, Kdm2 [Histones], bin3, RhoGAP93B, Gpdh3, Sara [BMP_TGFbeta], Plc21C, Su(z)2, mtd, ed, DmelCG10226, DmelCG17211, bdg, Kank
148.0-150.0 kDa	23	0	1	Protein_kinases; Ubiquitin_proteasome	routine	DmelCG7376 [Ubiquitin_proteasome], lh, Myo95E, Plas, CG17338, Alms1a, XNP, DmelCG13131, Vps15 [Protein_kinases], Ndg, gry, Ge-1, Cytip, CG43427, Pex23, cnn, MED1, S13b1
150.0-152.0 kDa	27	0	0	Notch; Protein_kinases; Ubiquitin_proteasome	routine	DmelCG11898, FER [Protein_kinases], SP2353, salm, Nost, BG-DS03192.2, DmelCG4294, Ser [Notch], rk, Rab3-GAP, Kpc1 [Ubiquitin_proteasome], DmelCG14120, EP(2)2054, fld, DmelCG8492, Eip75B, wisp, Corin
152.0-154.0 kDa	28	0	1	Autophagy; Hedgehog; Protein_kinases; Protein_phosphatases; Toll_lmd_NFkB	elevated	DmelCG4301, rad50, Atg17 [Autophagy], Nlg4, Dme_CG7627, rha, DmelCG10132, SNF4Agamma [Protein_kinases], ATP8A, Dred, drosha, Hipk [Protein_kinases], PTP-ER, I(2)03659, ci [Hedgehog], Tollo [Toll_lmd_NFkB], Oseg6, CG2691
154.0-156.0 kDa	23	0	0		routine	CT23878, unc-13-4A, if, Cc2d2a, Taf3, CG17465, CG8154, nolo, Polr3A, DmelCG42326, 18w, Grip163, dtn, jing, rdog, Pld, Ask1, DmelCG10211
156.0-158.0 kDa	22	0	0	Ubiquitin_proteasome	routine	tim, DmelCG10254, DmelCG43795, PolrMT, wde, Rlf1, Hecw [Ubiquitin_proteasome], Ctr, Br140, Muc68D, bsf, rho-5, lce1, Cap-D2, Girdin, pum, thr, Cemp-C
158.0-160.0 kDa	28	0	1	Protein_kinases	routine	lqfR, fred, Sik3 [Protein_kinases], Cont, CG7457, Tep2, mus101, Glu11, Tomosyn, Hmr, sle, Gmap, Tppil, svr, Chs2, CG11846, NSD, TANGO1
160.0-162.0 kDa	17	0	1	Protein_kinases; Toll_lmd_NFkB	routine	Ptp52F, rdgA, Hcf, Nup160, Rok [Protein_kinases], Mnm, sws, clu, FAM21, Pde11, Toll-7 [Toll_lmd_NFkB], Cg8945, LRR, CG6740, CG11578, ph-d, LarpaB
162.0-164.0 kDa	21	0	0	Protein_kinases	routine	CG16868, Tik [Protein_kinases], I(2)05819, DmelCG17233, DAAM, msn [Protein_kinases], mammo, mtt, mmd, Hr4, Pde9, NAT1, I(3)mbt, CG7701, tinc, pns, prc, DmelCG34347
164.0-166.0 kDa	22	0	1		routine	anne, tn, Top2, DmelCG16711, Lpt, sprn-E, Cpsf160, Usp10, mesh, Skel, BubR1, tok, Claspin, MED26, DmelCG7806, chb, Dark, sina
166.0-168.0 kDa	16	0	1	Protein_phosphatases	routine	CG9137, Blm, robo2, Dm_2L-23546, CG33291, Fak, MED23, mam, ph-p, loco, CG11005, Ptp69D [Protein_phosphatases], Fancd2, DmelCG12105, cv-d, rempa
168.0-170.0 kDa	23	0	0		routine	dtr, tutt, sns, DmelCG42788, Rbcn-3B, SoYb, CG18419, sli, sba, Tep4, fra, Kul, CG15028, DmelCG32681, DmelCG15894, Fancm, OtopLc, nsl1
170.0-172.0 kDa	15	0	1	Protein_kinases; Toll_lmd_NFkB	routine	Golgin245, Toll-6 [Toll_lmd_NFkB], Ank, zld, Acl, DmelCG42637, gyc76C, Pxn, CG13980, DmelCG15080, DmelCG4744, cnk, Abi [Protein_kinases], GEFmeso, PDZ-GEF
172.0-174.0 kDa	17	0	1	Ubiquitin_proteasome	routine	C3G, mahj, akk, DmelCG6454, NP_649372, Ulp1, MED14, salsa, Cadps, DmelCG31140, Shrm, Tmem131, raskol, MRP, Gyf, Usp47 [Ubiquitin_proteasome], CG4093

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
174.0-176.0 kDa	15	0	0		routine	DmelCG34027, Alh, RhoGAP15B, ninaC, Col4a1, sol, DmelCG33235, Uggt, plx, Nup214, chas, nompA, DmelCG8086, osp, RecQ4
176.0-178.0 kDa	12	0	1		routine	Ela6, DmelCG42390, lml1, ImpE1, frac, raptor, Pvr, Mekk1, CG13812, DmelCG11970, Duox, Sos
178.0-180.0 kDa	11	0	0	Protein_kinases	routine	DmelCG14322, larp, wge, CT39634, wake, CG32778, RhoGAPp190, Gcn2 [Protein_kinases], dec, CG13632, Asx
180.0-182.0 kDa	7	0	1	Notch	routine	lilli, sha, sno [Notch], ms(2)34Fe, side-III, Arms, Zmynd8
182.0-184.0 kDa	14	0	0		routine	LaB2, fliz, Eato, kkv, CG11620, fwd, Bap170, Agl, Fur2, Hasp, G9a, Patronin, Daxx, cmb
184.0-186.0 kDa	14	0	0		routine	cora, Q9VHT6_DROME, pcm, Gucy2d, pros, DmelCG17687, brn, wapl, sas, DmelCG8908, PolR1A, Alk, Srm234, DmelCG42685
186.0-188.0 kDa	11	0	0	Protein_kinases	routine	gek [Protein_kinases], DmelCG10543, Cad99C, mon2, Crag, lno80, Arip4, CG15703, arr, Cirt, Sec71
188.0-190.0 kDa	17	0	0	Histones	routine	Ac13E, Set1 [Histones], tho2, mfr, enc, l(1)G0196, woc, form3, Ac78C, CLIP-190, chif, unc-104, GluProRS, polybromo, hwt, AdamTS-A, CG32689
190.0-192.0 kDa	11	0	0		routine	DmelCG6040, DmelCG1801, DmelCG31635, Mrtf, Setx, Muc30E, gukh, anon-WO0140519.52, Chc, Gapvd1, Tdg
192.0-194.0 kDa	11	0	0		routine	Slik, nompC, DmelCG7029, ssp3, Abca3, c11.1, spri, ec, ATP8B, CG8473, vkg
194.0-196.0 kDa	14	0	0	Ubiquitin_proteasome	routine	Rbp, DIP2, DmelCG1494, ana1, Mur2B, Neur14 [Ubiquitin_proteasome], Trf2, DmelCG6052, CG7696, Muc68E, cic, Oseg2, DmelCG10185, mute
196.0-198.0 kDa	5	0	0		routine	Nup153, Zcchc7, THADA, mxc, Dcr-2
198.0-200.0 kDa	9	0	0	Ubiquitin_proteasome	routine	RhoGAP100F, LaB1, Remo, Ltn1 [Ubiquitin_proteasome], CG43867, Prosap, Usp32 [Ubiquitin_proteasome], Nrx-1, tyf
200.0-202.0 kDa	8	0	0	Histones	routine	Ptp4E, mop, srib, CadN2, Spt20, Cad74A, CG12973, gpp [Histones]
202.0-204.0 kDa	8	0	0	Notch	routine	velo, Mcr, stum, BG-DS03323.1 [Notch], CG40500, rsh, CG12436, Kdm5
204.0-206.0 kDa	11	0	0		routine	gig, CdGAPr, rols, fab1, hlk, ZAP3, SLO2, fs(1)h, CG15607, peb, CG11460
206.0-208.0 kDa	4	0	0		routine	CG42286, DmelCG13991, clos, fs(1)M3
208.0-210.0 kDa	11	0	0	Ubiquitin_proteasome	routine	didum, Spg11, Ubr1 [Ubiquitin_proteasome], Spt6, DmelCG15040, vir, Polr2A, anon-WO0140519.124, f, Gp210, Dlg5
210.0-212.0 kDa	12	0	0	Insulin_P13K_AKT_TOR; Protein_kinases	routine	Nup98-96, CG34048, PI3K68D [Insulin_P13K_AKT_TOR], DmelCG16779, Pura, DmelCG2861, eiF4G1, qin, CG34384, sti [Protein_kinases], tea, Chd1
212.0-214.0 kDa	12	0	0	Ubiquitin_proteasome	routine	CG8858 [Ubiquitin_proteasome], cac, CG9487, e(y)3, Pcf11, Cngl, Hel89B, Gug, DmelCG18304, Nup188, hang, px
214.0-216.0 kDa	7	0	0	Autophagy; Insulin_P13K_AKT_TOR	routine	PAPLA1, rictor [Insulin_P13K_AKT_TOR], RhoGEF64C, Atg2 [Autophagy], jr, tai, l(3)psg2
216.0-218.0 kDa	7	0	0		routine	BcDNA:GH04922, CG5521, Dscam4, Cad87A, Wdr81, sdt, ca
218.0-220.0 kDa	6	0	0		routine	PlexA, mim, Khc-73, mld, DmelCG13917, yl
220.0-222.0 kDa	8	0	0		routine	FBgn, Cad86C, garz, Sin3A, Stacl, Cda5, Cad88C, cana
222.0-224.0 kDa	9	0	0	Actin_tubulin; Protein_phosphatases	routine	Dscam1, pps, DmelCG14200, Ptp10D [Protein_phosphatases], Spc105R, mtg0, t-Grip91 [Actin_tubulin], Dora, Camta
224.0-226.0 kDa	13	0	0	Protein_kinases	routine	Mi-2, ldd, Mhc, Mur89F, Exn, Sbf, DmelCG15824, dop [Protein_kinases], Trpm, ana3, Lrp4, mspss, CG10020
226.0-228.0 kDa	8	0	0	RAS_ERK_MAPK	routine	Rab3-GEF [RAS_ERK_MAPK], eiF4G2, cno, shtd, Tmc, Dscam2, DmelCG5937, Cep290
228.0-230.0 kDa	5	0	0	Protein_phosphatases	routine	Dscam3, E(Pc), Lar [Protein_phosphatases], CG6947, PolQ
230.0-232.0 kDa	6	0	0		routine	asp, Helz, mbc, sif, X11Lbeta, PlexB
232.0-234.0 kDa	9	0	0		routine	Gits, RhoGAP19D, CG31652, Root, DmelCG10011, sbb, Spn, crb, ct
234.0-236.0 kDa	10	0	0		routine	eys, toc, Nirvana, Nav, Zir, DmelCG43366, lntS1, Nup205, Elys, nocte
236.0-238.0 kDa	6	0	0		routine	zip, Nipped-B, EP2317, DmelCG12499, Heatr1, Heatr5
238.0-240.0 kDa	8	0	1	Insulin_P13K_AKT_TOR	routine	Zasp52, fs(1)N, cher, Taf1, para, xmas, rod, lnr [Insulin_P13K_AKT_TOR]
240.0-242.0 kDa	6	0	0		routine	PolZ1, Myo81F, MESR4, axo, CG6448, ocm
242.0-244.0 kDa	4	0	0		routine	Pi4KIIalpha, Dab, Myo28B1, CG12432
244.0-246.0 kDa	6	0	0		routine	CG15207, Brr2, nrm, Mhd, klar, Lmpt
246.0-248.0 kDa	6	0	0	Histones; Ubiquitin_proteasome	routine	sdk, ash1 [Histones], r, Ubr3 [Ubiquitin_proteasome], na, spg
248.0-250.0 kDa	5	0	0		routine	Sur, zye, BRWD3, Adcy1, obe
250.0-252.0 kDa	4	0	0		routine	east, ck, Cad89D, cmet
252.0-254.0 kDa	3	0	0		routine	Jarid2, DmelCG2926, Sptz
254.0-256.0 kDa	4	0	0	Histones	routine	enok [Histones], brp, Dcr-1, cv-c
256.0-258.0 kDa	7	0	0	Histones	routine	Sec16, FBgn0053978, tay, PolE1, Set2 [Histones], trio, Tig
258.0-260.0 kDa	3	0	0	Histones	routine	Ch10, htk, tr [Histones]
260.0-262.0 kDa	2	0	0		routine	Ziz, Apc
262.0-264.0 kDa	5	0	0		routine	Mgtor, Ptip, hob, Wdr62, pyd
264.0-266.0 kDa	3	0	0		routine	Ncoa6, beta-Spec, FASN2
266.0-268.0 kDa	2	0	0		routine	CG6108, FASN3
268.0-270.0 kDa	2	0	0		routine	DmelCG7518, anon-WO03042407.98
270.0-272.0 kDa	3	0	0	Histones	routine	CG6915, kon, Hers [Histones]
272.0-274.0 kDa	3	0	0	Protein_kinases	routine	Rme-8, Wnk [Protein_kinases], Pzl
276.0-278.0 kDa	3	0	0		routine	shn, CG14318, FASN1
278.0-280.0 kDa	6	0	0	Protein_kinases	routine	alpha-Spec, Acc, Lrk [Protein_kinases], Fhos, kto, Prp8
280.0-282.0 kDa	5	0	0	Protein_kinases	routine	skd, Epg5, RhoGEF2, mTor [Protein_kinases], Not1
284.0-286.0 kDa	4	0	0		routine	osa, CG9817, tud, CG17926
286.0-288.0 kDa	2	0	0		routine	sev, bbg
288.0-290.0 kDa	4	0	0	Notch; Protein_kinases	routine	CAP, N [Notch], mei-41 [Protein_kinases], Piezo
290.0-292.0 kDa	1	0	0		routine	CG15099
292.0-294.0 kDa	4	0	0		routine	ndl, BEST-LD19244, Ca-alpha1D, l(3)80Fj
294.0-296.0 kDa	1	0	0		routine	mud
296.0-298.0 kDa	1	0	0		routine	PsGEF
298.0-300.0 kDa	3	0	0		routine	Nup358, Rim, tnc
300.0-302.0 kDa	2	0	0		routine	E(bx), CG11008
302.0-304.0 kDa	2	0	0	Ubiquitin_proteasome	routine	Ufd4 [Ubiquitin_proteasome], Nuak
304.0-306.0 kDa	1	0	0		routine	Ten-m
306.0-308.0 kDa	2	0	0		routine	Tet, rhea
308.0-310.0 kDa	3	0	0		routine	teq, Muc68Ca, CG1295
310.0-312.0 kDa	2	0	0		routine	faf, DmelCG4554
312.0-314.0 kDa	1	0	0		routine	Ppn
314.0-316.0 kDa	1	0	0		routine	Iva
316.0-318.0 kDa	2	0	0	Protein_kinases	routine	Nf1, tefu [Protein_kinases]
318.0-320.0 kDa	3	0	0	Ubiquitin_proteasome	routine	hyd [Ubiquitin_proteasome], Itpr, DmelCG14692
320.0-322.0 kDa	4	0	0	Ubiquitin_proteasome	routine	LUBEL [Ubiquitin_proteasome], NaCP60E, DmelCG13492, Bicra
322.0-324.0 kDa	2	0	0		routine	jbug, Megf8
326.0-328.0 kDa	2	0	0		routine	Plp, Ten-a
332.0-334.0 kDa	2	0	0		routine	zfh2, unc79
336.0-338.0 kDa	3	0	0	Histones; Ubiquitin_proteasome	routine	upSET [Histones], ctrip [Ubiquitin_proteasome], tou
338.0-340.0 kDa	1	0	0		routine	Muc12Ea
340.0-342.0 kDa	1	0	0	Histones	routine	nej [Histones]
342.0-344.0 kDa	1	0	0		routine	kmr

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
346.0-348.0 kDa	3	0	0		routine	M7BP, CadN, mo
348.0-350.0 kDa	1	0	0		routine	Myo10A
350.0-352.0 kDa	2	0	0		routine	dom, bark
354.0-356.0 kDa	2	0	0		routine	Ca-alpha1T, Su(var)2-HP2
356.0-358.0 kDa	1	0	0		routine	dikar
360.0-362.0 kDa	3	0	0	Protein_kinases	routine	unc-13, blobby, nonC [Protein_kinases]
364.0-366.0 kDa	1	0	0		routine	pcx
366.0-368.0 kDa	1	0	0		routine	unc80
370.0-372.0 kDa	1	0	0		routine	sov
372.0-374.0 kDa	1	0	0		routine	apolpp
374.0-376.0 kDa	2	0	0		routine	Vps13, wb
376.0-378.0 kDa	1	0	0		routine	Rbcn-3A
378.0-380.0 kDa	2	0	0		routine	clone, Smr
384.0-386.0 kDa	5	0	0		routine	RhoGEF3, fry, rg, Dmel[CG11122, ds
388.0-390.0 kDa	1	0	0		routine	bchs
390.0-392.0 kDa	1	0	0		routine	uif
394.0-396.0 kDa	1	0	0		routine	htt
396.0-398.0 kDa	1	0	0		routine	stan
398.0-400.0 kDa	1	0	0		routine	mv
400.0-402.0 kDa	1	0	0	Histones	routine	trx [Histones]
402.0-404.0 kDa	1	0	0		routine	Cubn2
404.0-406.0 kDa	1	0	0		routine	btsz
410.0-412.0 kDa	2	0	0		routine	Dys, LanA
412.0-414.0 kDa	1	0	0		routine	zormin
418.0-420.0 kDa	1	0	0		routine	Cubn
422.0-424.0 kDa	1	0	0		routine	mask
424.0-426.0 kDa	1	0	0		routine	Vps13B
426.0-428.0 kDa	1	0	0		routine	Hml
434.0-436.0 kDa	2	0	0		routine	Dmel[CG10631, Nipped-A
440.0-442.0 kDa	1	0	0	Ubiquitin_proteasome	routine	puf [Ubiquitin_proteasome]
444.0-446.0 kDa	1	0	0		routine	Vps13D
454.0-456.0 kDa	1	0	0		routine	Dhc62B
460.0-462.0 kDa	1	0	0		routine	Sdb
466.0-468.0 kDa	1	0	0		routine	Dhc36C
468.0-470.0 kDa	1	0	0		routine	Dhc16F
472.0-474.0 kDa	1	0	0		routine	Cdep
474.0-476.0 kDa	1	0	0		routine	Obse
480.0-482.0 kDa	1	0	0		routine	btv
494.0-496.0 kDa	1	0	0		routine	Apoltp
496.0-498.0 kDa	2	0	0		routine	Svil, kst
498.0-500.0 kDa	1	0	0		routine	trol
504.0-506.0 kDa	2	0	0		routine	Dnah3, Cht6
514.0-516.0 kDa	1	0	0		routine	Dhc93AB
522.0-524.0 kDa	1	0	0		routine	kug
524.0-526.0 kDa	1	0	0	Actin_tubulin	routine	Mical [Actin_tubulin]
528.0-530.0 kDa	1	0	0	Ubiquitin_proteasome	routine	HERC2 [Ubiquitin_proteasome]
530.0-532.0 kDa	3	0	0		routine	mgl, Dhc64C, LRP1
540.0-542.0 kDa	1	0	0		routine	Dhc1
550.0-552.0 kDa	1	0	0	Ubiquitin_proteasome	routine	Bruce [Ubiquitin_proteasome]
552.0-554.0 kDa	1	0	0		routine	Dmel[CG3339
556.0-558.0 kDa	1	0	0	Ubiquitin_proteasome	routine	HUWE1 [Ubiquitin_proteasome]
564.0-566.0 kDa	2	0	0	Ubiquitin_proteasome	routine	ft, hiw [Ubiquitin_proteasome]
572.0-574.0 kDa	1	0	0		routine	tweek
580.0-582.0 kDa	3	0	0		routine	CG3950, RyR, Dhc98D
590.0-592.0 kDa	1	0	0		routine	poe
592.0-594.0 kDa	2	0	0		routine	futsch, kis
598.0-600.0 kDa	1	0	0		routine	spen
630.0-632.0 kDa	1	0	0		routine	Dmel[CG13185
922.0-924.0 kDa	1	0	0		routine	Strn-Mlck
988.0-990.0 kDa	1	0	0		routine	shot
1000.0-1002.0 kDa	1	0	0		routine	bt
1404.0-1406.0 kDa	1	0	0		routine	Msp300
1488.0-1490.0 kDa	1	0	0		routine	Ank2
1768.0-1770.0 kDa	1	0	0		routine	Muc14A
2064.0-2066.0 kDa	1	0	0		routine	sls
2442.0-2444.0 kDa	1	0	0		routine	dpy

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
1.23	tal-1A	C0HJX4	Peptide tarsal-less 1A	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.23	tal-2A	C0HJX5	Peptide tarsal-less 2A	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.24	tal-3A	A3RLR0	Peptide tarsal-less 3A	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.72	DmelCG43739	M9PBF3	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.86	CG1461-5'utr	A0A1Z1CH01	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.89	CG6424-5'utr	A0A384SX97	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.98	DmelCG46270	A0A126GUS8	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.15	CG14253-5'utr	A0A1Z1CH02	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.30	DmelCG43178	A0A0B4LF48	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.31	DmelCG46525	A0A9F2H0Z1	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.36	DmelCG45088	A0A0B4LFA8	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.37	MSAmiP	A0ACD4DAS6	MSA micropeptide, isoform B	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.41	DmelCG43200	A0A0B4K848	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.47	noncoding_5744	A0A0B4LEP6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.51	MIP03715_ORF_1	A0A126GUN2	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.62	Acp98AB	O46201	Accessory gland protein 98AB	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.63	MIP03715	A0A0B4LG22	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.64	CG13627-5'utr	A0A1Z1CK60	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.64	DmelCG46335	A0A2U8U1T7	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.66	CG43200-3'utr	A0A384TTL1	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.72	Mst40	Q24437	Protein male-specific 40	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.77	DmelCG46310	A0A1Z1CH08	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.88	DmelCG45307	X2JDL1	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.88	CR33939	A0A0S0X126	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.97	CG7708-5'utr	A0A1Z1CH22	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.05	CG42784-5'utr	A0A1W5PW65	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.08	DmelCG46244	A0A0S0WGX5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.15	DmelCG42852	M9MS52	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.16	GH14785	A0A6F7UCJ3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.19	DmelCG44388	A0A384SYE0	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.25	SciA	C0HJH4	Sarcolamban A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.26	DmelCG45087	A0A0B4LFN2	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.29	crc-5'utr	A0A1W5Q018	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.40	RpL41	A1ZBY7	Ribosomal protein L41	2.0-4.0 kDa	Ribosomal_proteins	configured audit family match: Ribosomal_proteins
3.41	SciB	C0HJH3	Sarcolamban B	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.58	DmelCG46468	A0A6H2EJ23	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.59	DmelCG43362	M9NF92	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.65	Sfp96F	E1J1Y3	Seminal fluid protein 96F	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.70	CR33947	A0A0S0WJP9	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.70	Sfp79B	E1J1Z8	Seminal fluid protein 79	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.71	Dromel_CG17759_ga lpha9b_uORF	A0A0B4LG97	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.73	lincRNA.182	M9NET3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.73	DmelCG46458	A0A6H2EFK9	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.73	DmelCG46462	A0A6M3Q933	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.75	DmelCG46523	A0A9F2H0X9	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.78	DmelCG43221	A0A1W5PZ55	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.78	DmelCG46435	A0A6M3QGN8	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.80	DmelCG43732	A0A0B4KGG3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.85	MtnA	P04357	Metallothionein-1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.86	tal-AA	A3RLR1	Peptide tarsal-less AA	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.87	CG13004-5'utr	A0A1Z1CLY5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.87	DmelCG43220	M9PBD0	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.91	DmelCG43813	M9PCS7	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.92	CR44215	A0A1Z1CH16	Seminal fluid protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.99	MtnE	A0A0B4LHG6	Metallothionein E, isoform A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.99	Sfp36F	X2JEJ6	Seminal fluid protein 36F, isoform A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
4.04	DmelCG43965	M9PEK4	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.06	chr2L_11663230_116 63906	X2J9G7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.10	BomS3	Q9V8G0	Bomanin Short 3	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.15	DmelCG45099	A0A0B4LIG4	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.18	DmelCG43814	M9PDC8	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.18	Mst57Db	Q9VBL7	Accessory gland-specific peptide 57Db	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.19	BomS6	A1ZB64	Bomanin Short 6	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.20	gdl-ORF39	Q9U5V6	Gonadal protein gdl-ORF39	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.22	DmelCG43251	M9NDX2	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.26	BomS5	A1ZB63	Bomanin Short 5	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.30	lincRNA.174	M9NF77	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.31	DmelCG43645	A0A1W5PXS3	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.37	Ost4	A1Z7U3	Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit Ost4	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.38	Hdac3-5'utr	A0A0B4LGP2	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.43	DmelCG43390	M9NE78	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.52	Dso1	P82705	Daisho1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.53	MtnB	P11956	Metallothionein-2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.54	DmelCG43694	A0A0B4KGU1	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.55	MtnD	Q8I9B4	Metallothionein-4	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.57	DmelCG42781	M9MSA8	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.58	DmelCG43348	M9PBM9	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
4.59	DmelCG43397	A0A0B4K7L9	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.60	MtnC	Q9VDN2	Metallothionein-3	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.63	DmelCG43407	M9NG60	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.66	DmelCG33946	A0A0S0WGZ3	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.67	BomS1	P82706	Bomanin Short 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.70	BomS4	A0ACM8PYK4	Bomanin short 4	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.72	DmelCG43403	M9NCQ5	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.74	BomS2	O77150	Bomanin Short 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.74	DmelCG46311	A0A1Z1CGY8	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.80	DmelCG44439	A0A0B4LEX3	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.81	DmelCG43935	A0A0B4KI42	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.82	IBIN	A0A6M3Q7I6	Induced by infection	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.86	DmelCG44174	A0A0B4KGB6	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.91	DmelCG46059	A0A0S0WGU5	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.93	lncRNA-CR45045	A0A6H2EDS4	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.97	DmelCG43147	M9NFS2	Seminal fluid protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.98	DmelCG46309	A0A1W5Q0T5	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.99	Dso2	P83869	Daisho2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.01	DmelCG46464	A0A6H2EDJ9	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.05	DmelCG45690	X2JA25	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.07	DmelCG46307	A0A1W5PW68	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.07	DmelCG34212	A8DY47	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.08	DmelCG46426	A0A6H2EHD4	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.08	NEST-bs15d07	Q8MLN5	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.09	Neb-cGP	Q9W2U8	Neb-cGP, isoform A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.12	DMG8-chr2L_34.001	M9PD55	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.12	DmelCG42587	E1JHI4	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.13	CG33254	Q7KUV4	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.15	DmelCG43680	M9PFI7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.15	DmelCG42305	B7YZX3	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.17	Acp54A1	Q45WI1	Accessory gland protein 54A1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.17	DmelCG45084	X2JAK7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.18	noncoding_5956	A0A0B4K853	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.19	DmelCG46448	A0A6H2EI14	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.21	Dromel CG8026_FBI r0088611_uORF	A0A0B4LEZ9	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.22	CG8788-3'utr	A0A0B4LEI2	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.23	Mst84Dc	Q01644	Male-specific sperm protein Mst84Dc	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.23	Mst87F	P08175	Male-specific sperm protein Mst87F	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.26	DmelCG34173	A8DZ22	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.27	DmelCG42662	A0A0B4JD61	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.28	DmelCG43450	A0A0B4K671	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.28	Acp33A	O46227	Accessory gland peptide Acp33A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.30	DmelCG44257	A0A0B4LH88	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
DROS_Lizcano_BJ2003_dPKB	dPKB/Akt and phospho-dPKB western blots	58.0-60.0 kDa	Retraction lists cross-figure reuse involving dPKB, including Fig. 1B dPKB appearing as part of Fig. 7A dS6K, Fig. 2B dS6K appearing as Fig. 2B dPKB after horizontal transformation, Kc167 dS6K appearing as S2 dPKB after horizontal transformation, and Kc167 p-dPKB appearing in part as S2 p-dPKB after horizontal transformation.	A 2025 Biochemical Journal retraction documents multiple transformed or reused Drosophila dPKB Western-blot images, providing a direct audit precedent around the approximately 60 kDa Akt neighborhood.
DROS_Lizcano_BJ2003_dS6K	dS6K western blots	54.0-56.0 kDa	Retraction lists several dS6K similarities: Fig. 1B dPKB appears as part of Fig. 7A dS6K; Fig. 2B dS6K appears as Fig. 2B dPKB with horizontal transformation; Fig. 4A Kc167 dS6K appears in part as S2 dPKB with horizontal transformation; and Fig. 4B Kc167 dS6K appears as S2 dS6K with horizontal transformation.	The same retraction documents repeated or horizontally transformed dS6K Western-blot material across figures and protein labels, providing a direct Drosophila audit precedent in the current-reference approximately 55 kDa S6k neighborhood.
DROS_Urrutia_SciRep2016_pMad	Mad phospho-western blots (pMadS212 and pMadCter)	50.0-52.0 kDa	Scientific Reports correction states that the pMadS212 blot in Fig. 4c was unintentionally swapped with the pMadCter blot from Fig. 4d.	A publisher correction documents an identity swap between two phospho-Mad Western blots in Drosophila cells, mapping the protein-level audit precedent to the approximately 50.5 kDa Mad-PA neighborhood.
DROS_Urrutia_SciRep2016_Flag MadAVA	Flag-Mad-AVA western blots	50.0-52.0 kDa	Scientific Reports correction states that Flag-Mad-AVA blots were unintentionally swapped between Supplementary Fig. 2c and 2d.	The same correction documents swapped Flag-Mad-AVA blots between supplementary panels, providing a direct Drosophila blot-identity audit precedent around the Mad molecular-weight neighborhood.
DROS_Basu_SciRep2016_H3K9me3	H3K9me3 western blot	14.0-16.0 kDa	Scientific Reports author correction states that the H3K9me3 Western-blot example was captured in the wrong orientation and displayed at 180-degree rotation; a biological replicate replaced the panel. The correction also explains that similarity between H3K9me3 and H3 blots arose from stripping and reprobing.	A Drosophila author correction documents a 180-degree orientation error in an H3K9me3 Western blot, supplying a lower-severity presentation-integrity precedent in the approximately 15.4 kDa histone H3 neighborhood.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	UP000000803_7227.fasta
Marker file	drosophila_marker_genes.tsv
Signaling-prefix file	drosophila_signaling_prefixes.tsv
Family-pattern file	drosophila_family_patterns.tsv
Species/report label	Drosophila melanogaster